

# GUIDE TO USING SIAM'S L<sup>A</sup>T<sub>E</sub>X STYLE\*

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**Abstract.** Documentation is given for use of the SIAM standard L<sup>A</sup>T<sub>E</sub>X and B<sup>I</sup>B<sup>T</sup>E<sub>X</sub> macros. Instructions and suggestions for compliance with SIAM style standards are also included. Familiarity with standard L<sup>A</sup>T<sub>E</sub>X commands is assumed.

**Key words.** L<sup>A</sup>T<sub>E</sub>X, B<sup>I</sup>B<sup>T</sup>E<sub>X</sub>, SIAM Journals, Documentation

**MSC codes.** 00A20, 00B10

**1. Introduction.** This file is documentation for the SIAM L<sup>A</sup>T<sub>E</sub>X style, including how to typeset the main document, the B<sup>I</sup>B<sup>T</sup>E<sub>X</sub> file, and any supplementary material. More information about SIAM's editorial style can be found in the style manual, available at [https://epubs.siam.org/pb-assets/files/SIAM\\_STYLE\\_GUIDE\\_2019-1635349464967.pdf](https://epubs.siam.org/pb-assets/files/SIAM_STYLE_GUIDE_2019-1635349464967.pdf). The major changes in the SIAM standard class are summarized in [Appendix A](#). The SIAM L<sup>A</sup>T<sub>E</sub>X files can be found at <https://www.siam.org/journals/auth-info.php>. The files that are distributed for the standard macros are given below.

- `siamart251216.cls` (required): Main SIAM standard L<sup>A</sup>T<sub>E</sub>X class file.
- `siamplain.bst` (required): Bibliographic style file for B<sup>I</sup>B<sup>T</sup>E<sub>X</sub>.
- `docsiamart.tex`: Produces this documentation.
- `references.bib`: B<sup>I</sup>B<sup>T</sup>E<sub>X</sub> database for this documentation and examples.
- `ex_article.tex`: Template for article.
- `ex_supplement.tex`: Template for supplement.
- `ex_shared.tex`: Template for shared information for article and supplement.

To use these files, put `siamart251216.cls` and `siamplain.bst` in the directory with your paper or, alternatively, into your L<sup>A</sup>T<sub>E</sub>X and B<sup>I</sup>B<sup>T</sup>E<sub>X</sub> paths, respectively. The outline of a SIAM L<sup>A</sup>T<sub>E</sub>X article is shown in [Example 1](#). Templates are provided and discussed in more detail in [section 12](#).

## Example 1: Document outline

```
\documentclass{siamart251216}
% Preamble consisting of:
% Packages and macro definitions go here.
% Define title, authors, headers here.
\begin{document}
\maketitle
% Other front matter goes here: abstract, keywords, MSC codes.
% Main body goes here.
% Appendices goes here (optional).
% Acknowledgements go here (optional).
% Bibliography goes here.
\end{document}
```

---

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**2. Class options.** Class options can be included in the bracketed argument of the command, separated by commas. The possible class options are:

- **review** — Recommended for submitting your manuscript to a SIAM journal. Adds line numbers as well as the statement “This manuscript is for review purposes only” to the bottom of each page.
- **final** — Turns off the black boxes that help authors identify lines that are too long. The final published version will have this option on.
- **supplement** — Specifies that the file is a supplement and not the main document, causing changes in the appearance of the title and numbering; see [section 11](#) for details.
- **hidelinks** — Turns off colors on hyperlinks; see [section 4](#). The hyperlinks still exist, but there is no color to differentiate them. The final published version will have this option on.

**3. Front matter.** The title and author parts are formatted using the standard `\title`, `\author`, and `\maketitle` commands as described in Lamport [9]. The title and author should be declared in the preamble. The title and author names are automatically converted to uppercase in the document. If there is more than one author, each additional author should be preceded by the `\and` command. The addresses and support acknowledgments are added via `\thanks`. Each author’s thanks should specify their address. The support acknowledgment should be put in the title thanks, unless specific support needs to be specified for individual authors, in which case it should follow the author address. The header for this file was produced by the code in [Example 2](#), including an example of a shared footnote. Each thanks produces a footnote, so the footnote of the second author is #3. The command `\headers{title}{authors}` command, with the title (possibly shortened to fit) and the authors’ names, creates the page headers, automatically converted to uppercase.

#### Example 2: Title and authors in preamble

```
\title{Guide to Using SIAM’s \LaTeX\ Style\thanks{Submitted to the editors
DATE.
\funding{Funding information goes here.}}}

\author{Dianne Doe\thanks{Imagination Corp., Chicago, IL
(\email{ddoe@imag.com}).}
\and Paul T. Frank\thanks{Department of Applied Math, Fictional
University, Boise, ID (\email{ptfrank@fictional.edu},
\email{jesmith@fictional.edu}).}
\and Jane E. Smith\footnotemark[3]}

% Custom SIAM macro to insert headers
\headers{Guide to Using SIAM’S \LaTeX\ Style}{Dianne Doe, Paul T. Frank,
and Jane E. Smith}
```

Following the author and title is the abstract, key words listing, and MSC codes, designated using the `abstract`, `keywords`, and `MSCcodes` environments. Authors are responsible for providing MSC codes which can be found on the AMS web site [2]. The abstract, keywords, and MSC codes for this document are specified in [Example 3](#).

### Example 3: Abstract, keywords, and MSC codes

```
\newcommand{\BibTeX}{\scshape Bib}\TeX\xspace} % <- Preamble
\begin{abstract}
Documentation is given for use of the SIAM standard \LaTeX\ and \BibTeX\
macros. Instructions and suggestions for compliance with SIAM style
standards are also included. Familiarity with standard \LaTeX\ commands is
assumed.
\end{abstract}

\begin{keywords}
\LaTeX, \BibTeX, SIAM Journals, Documentation
\end{keywords}

\begin{MSCcodes}
00A20, 00B10
\end{MSCcodes}
```

A more complete example, including a PDF supplement, that uses the included files `ex.article.tex`, `ex.supplement.tex`, and `ex.shared.tex` is discussed in [section 12](#). The example files can be used as a starting point for producing a document.

**4. Cross references and hyperlinks.** SIAM now supports cross references and hyperlinks via the `cleveref` and `hyperef` packages, which are loaded by the class file.

**4.1. Cleveref.** SIAM strongly recommends using the commands provided by the `cleveref` package for cross referencing. The package is automatically loaded and already customized to adhere to SIAM's style guidelines. To create a cross reference, use the command `\cref` (inside sentence) or `\Cref` (beginning of a sentence) in place of the object name and `\ref`. The `cleveref` package enhances L<sup>A</sup>T<sub>E</sub>X's cross-referencing features, allowing the format of cross references to be determined automatically according to the "type" of cross reference (equation, section, etc.) and the context in which the cross reference is used. So, the package *automatically* inserts the object name as well as the appropriate hyperlink; see [Example 4](#). It may require two L<sup>A</sup>T<sub>E</sub>X compilations for the references to show up correctly. Additional examples are shown in the sections below for equations, tables, figures, sections, etc.

**Example 4: Advantage of using cleveref**

The normal way to get a cross reference with a hyperlink requires a lot of typing: `\hyperref[thm:mvt]{Theorem~\ref*{thm:mvt}}`.

The `\texttt{cleveref}` package gets both the name and hyperlink automatically using a single macro: `\cref{thm:mvt}`.

It also handles multiple references with the same macro, such as `\cref{thm:mvt,fig:pgfplots,fig:testfig}`.

The normal way to get a cross reference with a hyperlink requires a lot of typing: [Theorem 6.1](#). The `cleveref` package gets both the name and hyperlink automatically using a single macro: [Theorem 6.1](#). It also handles multiple references with the same macro, such as [Theorem 6.1](#) and [Figures 1](#) and [2](#).

**4.2. Hyperref.** Hyperlinks are created with the `\href` and `\url` commands, as shown in [Example 5](#). SIAM has also defined the `\email` command, as shown in [Example 2](#). You can hide links (i.e., turn off link colors) with the `hidelinks` option.

**Example 5: Creating hyperlinks**

The `\href{https://www.siam.org}{SIAM homepage}` has general information. Note that the colored text will `\emph{not}` appear in the print version nor will the hyperlink be active, so the writer may want to specify the location explicitly instead by using `\url{https://www.siam.org}`.

The [SIAM homepage](#) has general information. Note that the colored text will *not* appear in the print version nor will the hyperlink be active, so the writer may want to specify the location explicitly instead by using [https://www.siam.org](#).

Note that homepage links via `\url` in the `\thanks` environment require special formatting for the tilde (`~`) character. The formatting is used in the template and shown in [Example 26](#).

**5. Math and equations.** Here we show some example equations, with numbering, and examples of referencing the equations. SIAM now includes the package `amsmath` by default, and we include some of its features as well, although the reader should consult the package user manual for further guidance [1, 5]. Several of the example are adapted from Mittlebach and Goossen’s guide to L<sup>A</sup>T<sub>E</sub>X [10].

[Example 6](#) is a straightforward example of inline mathematics equations that does not use any special packages or features.

**Example 6: Inline math**

The following shows an example of math in text:

Let  $S=[s_{ij}]$  ( $1\leq i,j\leq n$ ) be a  $(0,1,-1)$ -matrix of order  $n$ .

The following shows an example of math in text: Let  $S=[s_{ij}]$  ( $1\leq i,j\leq n$ ) be a  $(0,1,-1)$ -matrix of order  $n$ .

In [Example 7](#), we show the recommended method for getting blackboard fonts using the `amsfonts` package. This is not loaded by default and must be included in the preamble.

**Example 7: Blackboard math**

`\usepackage{amsfonts} % <- Preamble`

Blackboard bold characters, such as  $\mathbb{C}$  and  $\mathbb{R}$ , should be created with the `\texttt{amsfonts}` package, although this is not included by default.

Blackboard bold characters, such as  $\mathbb{C}$  and  $\mathbb{R}$ , should be created with the `amsfonts` package, although this is not included by default.

Example 8 shows the `smallmatrix` environment for an inline matrix from the `amsmath` package, which is included by default.

**Example 8: Inline matrix**

Matrices of no more than two rows appearing in text can be created as shown in the next example:

```
$B = \bigl[ \begin{smallmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{smallmatrix} \bigr] $.
```

Matrices of no more than two rows appearing in text can be created as shown in the next example:  $B = \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix}$ .

Bigger matrices can be rendered with environments from the `amsmath` package, such as `bmatrix` and `pmatrix` used in Example 9.

**Example 9: Creating matrices**

Display matrices can be rendered using environments from `\texttt{amsmath}`:

```
\begin{equation}\label{eq:matrices}
S=\begin{bmatrix}1&0\\0&0\end{bmatrix}
\quad\text{and}\quad
C=\begin{pmatrix}1&1&0\\1&1&0\\0&0&0\end{pmatrix}.
\end{equation}
\Cref{eq:matrices} shows some example matrices.
```

Display matrices can be rendered using environments from `amsmath`:

$$(5.1) \quad S = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \quad \text{and} \quad C = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Equation (5.1) shows some example matrices.

**Example 10** shows how to use the `\DeclareMathOperator` command from the `amsoptn` package to declare the `\Range` macro. (This example also uses the `braket` package for the `\set` macro, but this is not necessarily recommended by SIAM.)

#### Example 10: Declaring math operators

```
\usepackage{braket,amfonts,amsoptn} % <- Preamble
\DeclareMathOperator{\Range}{Range} % <- Preamble
An example of a math operator:
\begin{equation}\label{eq:range}
  \Range(A) = \set{ y \in \mathbb{R}^n \mid y = Ax }.
\end{equation}
```

An example of a math operator:

$$(5.2) \quad \text{Range}(A) = \{ y \in \mathbb{R}^n \mid y = Ax \}.$$

**Example 11** shows how to use the `align` environment from `amsmath` to easily align multiple equations.

#### Example 11: Aligned equations

```
\Cref{eq:a,eq:b,eq:c} show three aligned equations.
\begin{align}
  f &= g, \label{eq:a} \\
  f' &= g', \quad \text{and} \quad \label{eq:b} \\
  \mathcal{L}f &= \mathcal{L}g \label{eq:c}.
\end{align}
```

Equations (5.3)–(5.5) show three aligned equations.

$$(5.3) \quad f = g,$$

$$(5.4) \quad f' = g', \quad \text{and}$$

$$(5.5) \quad \mathcal{L}f = \mathcal{L}g.$$

Another way to number a set of equations is the `subequations` environment from `amsmath`, as shown in **Example 12**.

**Example 12: Subequations**

We calculate the Fréchet derivative of  $F$  as follows:

```
\begin{subequations}
\begin{align}
F'(U,V)(H,K)
&= \langle R(U,V), H \rangle V^T + U \langle \Sigma K^T - \\
&P(H \langle \Sigma V^T + U \langle \Sigma K^T \rangle \rangle \rangle \langle \Sigma V^T + U \langle \Sigma K^T \rangle \rangle \\
&= \langle R(U,V), H \rangle V^T + U \langle \Sigma K^T \rangle \langle \Sigma V^T + U \langle \Sigma K^T \rangle \rangle \\
&\quad \langle \Sigma V^T + U \langle \Sigma K^T \rangle \rangle \\
&= \langle R(U,V) V \langle \Sigma V^T + U \langle \Sigma K^T \rangle \rangle, H \rangle + \\
&\quad \langle \Sigma V^T + U \langle \Sigma K^T \rangle \rangle \langle R(U,V), K^T \rangle. \quad \text{\label{eq:bb}}
\end{align}
\end{subequations}
\text{\Cref{eq:aa}} is the first line, and \text{\cref{eq:bb}} is the last line.
```

We calculate the Fréchet derivative of  $F$  as follows:

$$(5.6a) \quad F'(U, V)(H, K) = \langle R(U, V), H \rangle V^T + U \langle \Sigma K^T - P(H \langle \Sigma V^T + U \langle \Sigma K^T \rangle \rangle) \rangle \\ = \langle R(U, V), H \rangle V^T + U \langle \Sigma K^T \rangle$$

$$(5.6b) \quad = \langle R(U, V) V \langle \Sigma V^T + U \langle \Sigma K^T \rangle \rangle, H \rangle + \langle \Sigma V^T + U \langle \Sigma K^T \rangle \rangle \langle R(U, V), K^T \rangle.$$

Equation (5.6a) is the first line, and (5.6b) is the last line.

For an equation split over multiple lines, Example 13 shows the usage of the `multline` environment provided by `amsmath`.

**Example 13: Equation split across lines**

We claim that the projection  $g(U, V)$  is given by the pair of matrices:

```
\begin{multline}
g(U, V) = \biggl( \frac{R(U, V) V \langle \Sigma V^T + U \langle \Sigma K^T \rangle \rangle}{2} U, \\
- U \langle \Sigma V^T + U \langle \Sigma K^T \rangle \rangle \langle R(U, V), K^T \rangle \\
\frac{R(U, V)^T U \langle \Sigma V^T + U \langle \Sigma K^T \rangle \rangle}{2} V \biggr).
\end{multline}
```

We claim that the projection  $g(U, V)$  is given by the pair of matrices:

$$(5.7) \quad g(U, V) = \left( \frac{R(U, V) V \langle \Sigma V^T + U \langle \Sigma K^T \rangle \rangle}{2} U, \right. \\ \left. \frac{R(U, V)^T U \langle \Sigma V^T + U \langle \Sigma K^T \rangle \rangle}{2} V \right).$$

**6. Theorem-like environments.** SIAM loads `ntheorem` package and uses it to define the following theorem-like environments: `theorem`, `lemma`, `corollary`, `definition`, and `proposition`. SIAM also defines a `proof` environment that automatically inserts the symbol “□” at the end of any proof, even if it ends in an equation environment. *Note that the document may need to be compiled twice for the mark to appear.* Some of the calculus examples were adapted from [4]. Example 14 shows usage of the `theorem` environment. An optional argument can be used to

name the theorem. [Example 15](#) illustrates a corollary, without a name, and the proof environment.

#### Example 14: Theorem

```
\begin{theorem}[Mean Value Theorem]\label{thm:mvt}
  Suppose  $f$  is a function that is continuous on the closed interval
   $[a,b]$  and differentiable on the open interval  $(a,b)$ .
  Then there exists a number  $c$  such that  $a < c < b$  and
  \begin{displaymath}
    f'(c) = \frac{f(b)-f(a)}{b-a}.
  \end{displaymath}
  In other words,  $f(b)-f(a) = f'(c)(b-a)$ .
\end{theorem}
```

THEOREM 6.1 (Mean Value Theorem). *Suppose  $f$  is a function that is continuous on the closed interval  $[a,b]$  and differentiable on the open interval  $(a,b)$ . Then there exists a number  $c$  such that  $a < c < b$  and*

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

*In other words,  $f(b) - f(a) = f'(c)(b - a)$ .*

#### Example 15: Corollary and proof

```
\begin{corollary}
  Let  $f(x)$  be continuous and differentiable everywhere. If  $f(x)$ 
  has at least two roots, then  $f'(x)$  must have at least one root.
\end{corollary}
\begin{proof}
  Let  $a$  and  $b$  be two distinct roots of  $f$ .
  By \cref{thm:mvt}, there exists a number  $c$  such that
  \begin{displaymath}
    f'(c) = \frac{f(b)-f(a)}{b-a} = \frac{0-0}{b-a} = 0.
  \end{displaymath}
\end{proof}
```

COROLLARY 6.2. *Let  $f(x)$  be continuous and differentiable everywhere. If  $f(x)$  has at least two roots, then  $f'(x)$  must have at least one root.*

*Proof.* Let  $a$  and  $b$  be two distinct roots of  $f$ . By [Theorem 6.1](#), there exists a number  $c$  such that

$$f'(c) = \frac{f(b) - f(a)}{b - a} = \frac{0 - 0}{b - a} = 0.$$

□

SIAM also defines commands to create your own theorem- and remark-like environments:

- `newsiamthm` — Small caps header, italicized body.
- `newsiamremark` — Italics header, roman body.

Each command takes two arguments. The first is the environment name, and the

second is the name to show in the document. These commands should be used instead of `\newtheorem`. Examples 16 and 17 shows how to use the commands above, including how to specify the plural version for `cleveref` if it is unusual.

NOTE: Due to a current limitation of `cleveref`, any custom defined theorem-like environment aside from the ones listed in the *SIAM Style Manual (Chapter 6)* will need to use `\ref` instead of `\cref` when referenced in the document. The theorem-like environments listed in the style manual that are safe to use with `cleveref` are `theorem`, `lemma`, `corollary`, `definition`, `proposition`, `fact`, `claim`, `conclusion`, `result`, `consequence`, and `problem`.

An additional note is that `fact`, `claim`, `conclusion`, `result`, `consequence`, and `problem` can also be defined as a remark-like environment in accordance with the *SIAM Style Manual (Chapter 6)* without having an issue with `cleveref` if an author chooses to do so.

#### Example 16: New theorem-like environment

```
\newsiamthm{claim}{Claim} % <- Preamble
\nnewsiamremark{hypothesis}{Hypothesis} % <- Preamble
\crefname{hypothesis}{Hypothesis}{Hypotheses} % <- Preamble
\begin{claim}\label{cl:constant}
  If  $f'(x) = 0$  for all  $x \in (a,b)$  then  $f(x)$  is constant on  $(a,b)$ .
\end{claim}
\begin{hypothesis}\label{hyp1}
  The function  $f$  is continuously differentiable.
\end{hypothesis}
\begin{hypothesis}\label{hyp2}
  The random variable is normally distributed.
\end{hypothesis}
```

CLAIM 6.3. If  $f'(x) = 0$  for all  $x \in (a,b)$  then  $f(x)$  is constant on  $(a,b)$ .

Hypothesis 6.4. The function  $f$  is continuously differentiable.

Hypothesis 6.5. The random variable is normally distributed.

#### Example 17: References

We can reference multiple types of objects with a single reference:  
`\cref{cl:constant,thm:mvt,hyp1,hyp2}`.

We can reference multiple types of objects with a single reference: Claim 6.3, Theorem 6.1, and Hypotheses 6.4 and 6.5.

**7. Tables.** Table captions should go above the tables. Example 18 shows the code to generate Table 1. A more complicated example is shown in Example 19, which generates Table 2. This example uses subfloats via the `subfig` package, as well as special column options from the `array` package.

## Example 18: Example table.

```
\begin{table}[tbhp]
\footnotesize
\caption{Example table}\label{tab:simpletable}
\begin{center}
\begin{tabular}{|c|c|c|} \hline
Species & \bf Mean & \bf Std.^Dev. \\ \hline
1 & 3.4 & 1.2 \\ \hline
2 & 5.4 & 0.6 \\ \hline
\end{tabular}
\end{center}
\end{table}
```

TABLE 1  
Example table

| Species | Mean | Std. Dev. |
|---------|------|-----------|
| 1       | 3.4  | 1.2       |
| 2       | 5.4  | 0.6       |

## Example 19: Example table with subtables.

```
\usepackage{array} % <- Preamble
\usepackage[caption=false]{subfig} % <- Preamble
\newcolumntype{R}{>{\$}r<{\$}} %
\newcolumntype{V}[1]{>{\$}R@{\$}R<{\$}} %
\begin{table}[tbhp]
\footnotesize
\captionsetup{position=top} %<- Needed for using subtables created with the subfig package
\caption{Example table adapted from Kolda and Mayo \rm{\cite{KoMa14}}}\label{tab:KoMa14}
\begin{center}
\subfloat[ $\beta=1$ ]{
\begin{tabular}{|r|R|V{3}|c|r@{\$}\pm{\$}\,|l|} \hline
occ. & \multicolumn{1}{c|}{ $\lambda$ } & \multicolumn{4}{c|}{ $\mathbf{x}$ } & \\ \hline
fevals & \multicolumn{2}{c|}{time (sec.)} & \multicolumn{4}{c|}{} \\ \hline
718 & 11.3476 & 0.5544 & 0.3155 & 1.2018 & 0.0977 & 45 & 0.17 & 0.06 \\ \hline
134 & 3.7394 & 0.2642 & -1.1056 & 0.2657 & -0.3160 & 31 & 0.12 & 0.05 \\ \hline
4 & \multicolumn{4}{c|}{\emph{--- Failed to converge ---}} & 0.21 & 0.10 \\ \hline
\end{tabular}
}
\subfloat[ $\beta=-1$ ]{
\begin{tabular}{|r|R|V{3}|c|r@{\$}\pm{\$}\,|l|} \hline
occ. & \multicolumn{1}{c|}{ $\lambda$ } & \multicolumn{4}{c|}{ $\mathbf{x}$ } & \\ \hline
fevals & \multicolumn{2}{c|}{time (sec.)} & \multicolumn{4}{c|}{} \\ \hline
72 & -1.1507 & 0.2291 & 0.6444 & 0.3540 & -0.8990 & 34 & 0.14 & 0.06 \\ \hline
624 & -6.3985 & 0.1003 & 0.1840 & 0.5305 & 1.2438 & 48 & 0.19 & 0.08 \\ \hline
2 & \multicolumn{4}{c|}{\emph{--- Failed to converge ---}} & 0.23 & 0.02 \\ \hline
\end{tabular}
}
\end{center}
\end{table}
```

**8. Figures.** It is recommended that all figures be generated in high resolution. In the past, SIAM has required encapsulated postscript (EPS) format for final production. This is still an acceptable format, but SIAM also now allows high-resolution PDF, JPEG, and PNG figures. If working with EPS images and using `pdflatex`, we recommend the package `epstopdf` to automatically convert EPS images to PDF for inclusion in PDF documents created by `pdflatex`. Please note that `epstopdf` requires `texlive-font-utils` which may not be part of a standard  $\text{\LaTeX}$  installation. Example 20 shows the code to generate Figure 1. This example uses the `graphicx`

TABLE 2  
Example table adapted from Kolda and Mayo [8].

| occ. | $\lambda$              | $\mathbf{x}$                      | fevals | time (sec.)     |
|------|------------------------|-----------------------------------|--------|-----------------|
| 718  | 11.3476                | [ 0.5544 0.3155 1.2018 0.0977 ]   | 45     | $0.17 \pm 0.06$ |
| 134  | 3.7394                 | [ 0.2642 -1.1056 0.2657 -0.3160 ] | 31     | $0.12 \pm 0.05$ |
| 4    | — Failed to converge — |                                   |        | $0.21 \pm 0.10$ |

(a)  $\beta = 1$ 

| occ. | $\lambda$              | $\mathbf{x}$                     | fevals | time (sec.)     |
|------|------------------------|----------------------------------|--------|-----------------|
| 72   | -1.1507                | [ 0.2291 0.6444 0.3540 -0.8990 ] | 34     | $0.14 \pm 0.06$ |
| 624  | -6.3985                | [ 0.1003 0.1840 0.5305 1.2438 ]  | 48     | $0.19 \pm 0.08$ |
| 2    | — Failed to converge — |                                  |        | $0.23 \pm 0.02$ |

(b)  $\beta = -1$ 

package for the `\includegraphics` command.

#### Example 20: Example figure with subfigures and external files

```
\usepackage{graphicx,epstopdf} % <- Preamble
\usepackage[caption=false]{subfig} % <- Preamble
\begin{figure}[tbhp]
  \centering
  \subfloat[ $\epsilon_{\max}=5$ ]{\label{fig:a}\includegraphics{example_fig1}}
  \subfloat[ $\epsilon_{\max}=0.5$ ]{\label{fig:b}\includegraphics{example_fig2}}
  \caption{Example figure using external image files.}
  \label{fig:testfig}
\end{figure}
```

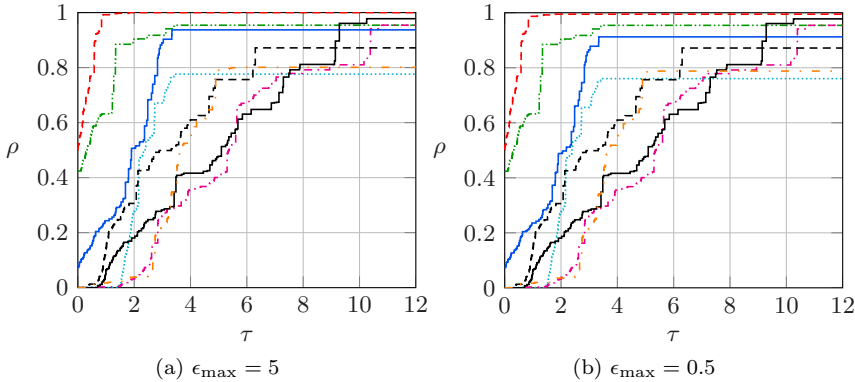


FIG. 1. Example figure using external image files.

Another option for figures is a graphics-generator that is platform- and format-independent. PGF is a TeX macro package for generating such graphics and works together with the most important TeX backend drivers, including pdftex and dvips. The user-friendly syntax layer called TikZ. Here we show an example using PGFPLOTS, useful for drawing high-quality plots directly in L<sup>A</sup>T<sub>E</sub>X. [Example 21](#) and [Example 22](#) shows the data and code, respectively, to generate [Figure 2](#), adapted from [6].

Example 21: Example data file (data.dat)

| d2_dof | d2_l2_err | d3_dof | d3_l2_err |
|--------|-----------|--------|-----------|
| 5      | 8.312e-02 | 7      | 8.472e-02 |
| 17     | 2.547e-02 | 31     | 3.044e-02 |
| 49     | 7.407e-03 | 111    | 1.022e-02 |
| 129    | 2.102e-03 | 351    | 3.303e-03 |
| 321    | 5.874e-04 | 1023   | 1.039e-03 |
| 769    | 1.623e-04 | 2815   | 3.196e-04 |
| 1793   | 4.442e-05 | 7423   | 9.658e-05 |
| 4097   | 1.207e-05 | 18943  | 2.873e-05 |
| 9217   | 3.261e-06 | 47103  | 8.437e-06 |

Example 22: Example TikZ/PGF for platform-independent graphics.

```

\usepackage{pgfplots} % <- Preamble
\begin{figure}[tbhp]
  \centering
  \begin{tikzpicture}
    \begin{loglogaxis}[height=2.75in, grid=major,
      xlabel={Degrees of Freedom}, ylabel={$L_2$ Error},
      legend entries={$d=2$, $d=3$}]
      \addplot table [x=d2_dof,y=d2_l2_err] {data.dat};
      \addplot table [x=d3_dof,y=d3_l2_err] {data.dat};
    \end{loglogaxis}
  \end{tikzpicture}
  \caption{Example \texttt{PGFPLOTS} figure.}
  \label{fig:pgfplots}
\end{figure}

```

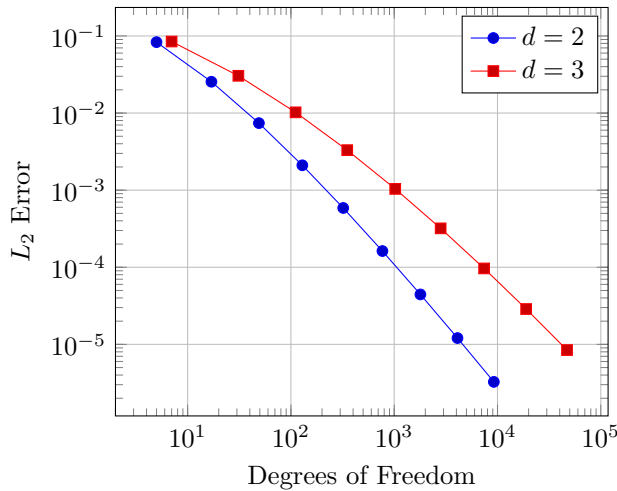


FIG. 2. Example PGFPLOTS figure.

**9. Algorithms.** SIAM automatically includes the `algorithm` package in the class definition. This provides the float environment. Users have the choice of `algpseudocode`, `algorithmic`, and other packages for actually formatting the algorithm. For example, Algorithm 9.1 is produced by the code in Example 23. In order to reference lines within the algorithm, we need to tell the `cleveref` package

how to do the referencing, which is the second line of [Example 23](#). Then we can use the code `\cref{line3}` to produce [Line 3](#).

### Example 23: Example algorithm

```
\usepackage{algorithmic} % <- Preamble
\crefname{ALC@unique}{Line}{Lines} % <- Preamble
\begin{algorithm}
\caption{Build tree}
\label{alg:buildtree}
\begin{algorithmic}[1]
\STATE{Define  $P:=T:=\{\{1\}, \ldots, \{d\}\}$ }
\WHILE{ $\#P > 1$ }
\STATE{\label{line3}{Choose  $C^{\prime} \in \mathcal{C}_p(P)$  with  $C^{\prime} := \operatorname{argmin}_{C \in \mathcal{C}_p(P)} \varrho(C)$ }}
\STATE{Find an optimal partition tree  $T_{C^{\prime}}^{\prime}$ }
\STATE{Update  $P := (P \setminus C^{\prime}) \cup \{\bigcup_{t \in C^{\prime}} t\}$ }
\STATE{Update  $T := T \cup \{\bigcup_{t \in \tau} t : \tau \in T_{C^{\prime}}^{\prime} \setminus \mathcal{L}(T_{C^{\prime}}^{\prime})\}$ }
\ENDWHILE
\RETURN  $T$ 
\end{algorithmic}
\end{algorithm}
```

---

### Algorithm 9.1 Build tree

---

- 1: Define  $P := T := \{\{1\}, \dots, \{d\}\}$
  - 2: **while**  $\#P > 1$  **do**
  - 3:   Choose  $C' \in \mathcal{C}_p(P)$  with  $C' := \operatorname{argmin}_{C \in \mathcal{C}_p(P)} \varrho(C)$
  - 4:   Find an optimal partition tree  $T_{C'}$
  - 5:   Update  $P := (P \setminus C') \cup \{\bigcup_{t \in C'} t\}$
  - 6:   Update  $T := T \cup \{\bigcup_{t \in \tau} t : \tau \in T_{C'} \setminus \mathcal{L}(T_{C'})\}$
  - 7: **end while**
  - 8: **return**  $T$
- 

**10. Sections.** Sections are denoted using standard L<sup>A</sup>T<sub>E</sub>X section commands, i.e., `\section`, `\subsection`, etc. If you wish to end the section title with something other than a period (the default), you have to add the command `\nopunct` at the end of the title.

Appendices are created with the normal sectioning commands, following the command `\appendix`. Titles of appendices created with `\section` are preceded by the word “Appendix,” but not the subsections or appendices created with `\section*`. Unlike normal sections, appendix sections may be sensitive to blank lines following the declaration, causing a new paragraph rather than the text immediately following the appendix title. This can be corrected by removing and blank lines. Any numbered, labeled sections can be referenced using `\cref`, including those without a title. Section titles are automatically inserted into the table of contents and converted to bookmarks; see [Appendix B.6](#) for handling special characters.

The acknowledgments section comes immediately before the references and after any appendices. It should be declared by `\section*{Acknowledgments}`.

**11. Supplemental material.** For several SIAM journals, authors are encouraged to submit Supplementary Materials to complement their articles. This might include additional figures or examples, animations, data sets used in the paper, computer code used to generate figures or tables, or other materials that are necessary to

fully document the research contained in the paper or to facilitate the readers’ ability to understand and extend the work.

The class option `supplement` should be used in the supplemental L<sup>A</sup>T<sub>E</sub>X file provided for creating PDF supplemental material. The supplement should have the same title and authors as the main document. The title is modified automatically by the SIAM class file so that it is preceded by the text “Supplementary Materials”, followed by a colon. The numbering is modified so that all sections, equations, figures, tables, algorithms, and so on to start with “SM”. A supplement does have sections but does not have an abstract, keywords, MSC codes, or appendices. The main document and supplement can cross reference sections, equations, theorem-like declarations, figures, tables, algorithms, etc. However, there is no sharing of references. The references are optional for a supplement. A template is provide, as discussed in [section 12](#).

**12. Template.** The files `ex_article.tex`, `ex_shared.tex`, and `ex_supplement.tex` provide a template that can be used for creating a L<sup>A</sup>T<sub>E</sub>X document with an optional supplement. [Examples 24](#) and [25](#) give the outline of an article and the supplement. In this case we assume that the title and authors are defined in the `ex_shared.tex` file, shown in [Example 26](#). Cross referencing between the main document and the supplement is enabled using the `xr-hyperref` package (included by the SIAM class file). Use `\externaldocument` to specify the external document to search for external references.

#### Example 24: Document outline with supplement

```
\documentclass{siamart251216}
\input{ex_shared}
\externaldocument{ex_supplement}
\begin{document}
\maketitle
% Other front matter: abstract, keywords, subject classifications.
% Main body goes here.
% Appendices and/or acknowledgments.
% Bibliography
\end{document}
```

#### Example 25: Supplement document outline

```
\documentclass[supplement]{siamart251216}
\input{ex_shared}
\externaldocument{ex_article}
\begin{document}
\maketitle
% No abstract, keywords, subject classifications.
% Main body goes here.
% No appendices or acknowledgments.
% Optional bibliography goes here.
\end{document}
```

[Example 26](#) (from `ex_shared.tex`) shows how the “shared” title and authors may be defined across the main document and a supplement. Note the use of the `\string` command in the URL for the tilde; this is only necessary inside the `\thanks` command.

**Example 26: Example of shared title and author macros**

```
% Add a serial/Oxford comma by default.
\newcommand{\creflastconjunction}{, and~}

% Used for creating new theorem and remark environments
\newsiamremark{remark}{Remark}
\newsiamremark{hypothesis}{Hypothesis}
\crefname{hypothesis}{Hypothesis}{Hypotheses}
\newsiamthm{claim}{Claim}
\newsiamremark{fact}{Fact}
\crefname{fact}{Fact}{Facts}

% Sets running headers as well as PDF title and authors
\headers{An Example Article}{D. Doe, P. T. Frank, and J. E. Smith}

% Title. If the supplement option is on, then "Supplementary Material"
% is automatically inserted before the title.
\title{An Example Article\thanks{Submitted to the editors DATE.}
\funding{This work was funded by the Fog Research Institute under contract
no.~FRI-454.}}

% Authors: full names plus addresses.
\author{Dianne Doe\thanks{Imagination Corp., Chicago, IL
(\email{ddoe@imag.com}, \url{http://www.imag.com/\string~ddoe/}).}
```

**13. Bibliography.** The SIAM BIB<sub>T</sub>E<sub>X</sub> style file, now called `siamplain.bst`, has been updated to include the new keys listed below:

- `doi`: Digital object identifier, a unique alphanumeric string
- `url`: Web address, usually impermanent
- `urldate`: Date that the web address was last accessed
- `eprint`: Archive identifier, a unique alphanumeric string
- `eprintclass`: Archive class
- `archive`: Archive URL, defaults to `https://arXiv.org/abs`
- `archivepreprint`: Archive name, defaults to “arXiv”
- `eid`: Article ID, if there are no page numbers
- `pagetotal`: Total number of pages, for use with article ID

Every entry type has been modified to include an optional link to a DOI, a URL, and/or an archive preprint reference. Additionally, the `article` entry now supports an Article ID, `eid`, and number of pages, `pagetotal`. To use this, include the following code in your L<sup>A</sup>T<sub>E</sub>X source code: `\bibliographystyle{siamplain}`.

**13.1. DOI.** A digital object identifier (DOI) is a unique alphanumeric string that provides a persistent link to its location on the Internet. The publisher assigns a DOI when your article is published and made available electronically. Using the `doi` field in BIB<sub>T</sub>E<sub>X</sub> to specify it, as shown for [8] in [Example 27](#); observe the new `doi` field which produces a hyperlink in the citation. Do not include the full URL, i.e., `https://doi.org/` preceding the DOI. Authors are highly advised to provide DOIs; if they are not provided, they will be requested during the copyediting process.

**Example 27: Example article in BibTeX**

```
@Article{KoMa14,
  title =      {An Adaptive Shifted Power Method for Computing
                Generalized Tensor Eigenpairs},
  author =     {Tamara G. Kolda and Jackson R. Mayo},
  doi =        {10.1137/140951758},
  journal =    {SIAM Journal on Matrix Analysis and Applications},
  number =     4,
  volume =     35,
  year =       2014,
  month =      dec,
  pages =      {1563--1581},
}
```

**13.2. URL.** There is also now support for the `url` field. Generally, the DOI is preferred to the URL, since the DOIs should be permanent references. For that reason, it is good practice to specify the last date that the URL was accessed, which is specified by the optional `urldate` field. Reference [7] produced by [Example 28](#) shows an example of using these fields.

**Example 28: Example with the URL field in BibTeX**

```
@Misc{Hi14,
  author =      {Nick Higham},
  title =       {A Call for Better Indexes},
  howpublished = {SIAM Blogs},
  year =        2014,
  month =       nov,
  url =         {http://blogs.siam.org/a-call-for-better-indexes/},
  urldate =     {2015-04-05}
}
```

**13.3. Preprint servers such as arXiv.** More and more manuscripts are available on preprint servers. In fact, SIAM's publication policy explicitly allows the final accepted version of any article to be posted on a preprint server such as arXiv.

For an arXiv paper, the `eprint` field is used to specify the identifier. The optional `eprintclass` field specifies the class. [Example 29](#) shows the BibTeX for [12].

**Example 29: Example arXiv reference in BibTeX**

```
@Misc{PeKoPi14,
  title = {Accelerating Community Detection by Using {K}-core Subgraphs},
  author = {Chengbin Peng and Tamara G. Kolda and Ali Pinar},
  eprint = {1403.2226},
  year =   2014,
  month =  mar,
  eprintclass = {math.NA}
}
```

Other preprint servers are supported as well, but these require specification of the fields `archive` and `archiveprefix`. In this case, the target URL is formed by concatenating the `archive`, a forward slash (/), and the `eprint`; and the text for the hyperlink is formed by concatenating the `archiveprefix`, a colon (:), and the `eprint`. [Example 30](#) shows the code to generate [13], including the preprint from PubMed. Note that this example has both the journal citation as well as the link for the preprint.

#### Example 30: Example PubMed reference in BibT<sub>E</sub>X

```
@Article{WoZhMeSh05,
  author =      {Woessner, Donald E. and Zhang, Shanrong and
                  Merritt, Matthew E. and Sherry, A. Dean},
  title = {Numerical Solution of the {Bloch} Equations Provides Insights
           into the Optimum Design of {PARACEST} Agents for {MRI}},
  journal =     {Magnetic Resonance in Medicine},
  doi =         {10.1002/mrm.20408},
  volume =      53,
  number =      4,
  month =       apr,
  year =        2005,
  pages =       {790--799},
  archiveprefix = {PubMed},
  archive =     {https://www.ncbi.nlm.nih.gov/pubmed},
  eprint =      {15799055}
}
```

**13.4. Article ID.** Some journals use an article ID rather than page numbers. The field `eid` specifies the article ID. The optional field `pagetotal` can say the number of pages in the document. An example of an article using these fields is shown in [Example 31](#) for citation [11].

#### Example 31: Example article ID reference in BibT<sub>E</sub>X

```
@Article{Ne03,
  title =       {Properties of Highly Clustered Networks},
  author =      {Newman, M. E. J.},
  doi =         {10.1103/PhysRevE.68.026121},
  journal =     {Phys. Rev. E},
  volume =      {68},
  year =        {2003},
  eid =         {026121},
  pagetotal =   6,
  month =       aug,
}
```

**13.5. Software citations.** SIAM encourages software citations, both related technical publications as well as the software itself. A citation to a software package may look something like what is shown in [Example 32](#) for citation [3]. Notice the double braces around the `author` key; else, it would appear as “C. D. Team”.

**Example 32: Example software reference in BIB<sub>T</sub>E<sub>X</sub>**

```

@misc{clawpack,
  title =      {Clawpack Software},
  author =     {{Clawpack Development Team}},
  url =        {http://www.clawpack.org},
  urldate =    {2015/05/14},
  note =       {Version 5.2.2},
  year =       2015
}

```

**Appendix A. Summary of major changes.**

Here we summarize the major changes in the latest version of the SIAM standard L<sup>A</sup>T<sub>E</sub>X and BIB<sub>T</sub>E<sub>X</sub> classes:

- Change in file name: `siamart251104.cls` is replaced by `siamart251216.cls`.
- Additional margin adjustments for print vendors.
- Adjusted document trim to match print publication trim of 6.75" x 10".
- Added a fix for `cleveref` referencing anything defined as a theorem-like environment as a theorem.
- Added a fix for `\label` and `\refstepcounter` due to `hyperref` and `cleveref` package changes.
- Change from using "AMS subject classifications" to "MSC codes".
- Added a correction needed to fix a problem caused by a recent change to the `eqnarray` environment (see a discussion regarding this [here](#)).
- Change in file name: `siam.bst` bibliography style file is replaced by `siamplain.bst`; see [section 1](#).
- Hyperlinking in cross references via the `cleveref` package, including customizations to adhere to SIAM conventions. Automatic PDF bookmarks enabled for sections, appendices, and references. See [subsection 4.1](#).
- Colored hyperlinks (red for external, green for internal). Can be disabled with class options `hidelinks`; see [subsection 4.2](#).
- New fields for BIB<sub>T</sub>E<sub>X</sub>, as listed in [section 13](#).
- Support for supplemental PDF files, including cross references between the supplement and the main document; see [section 11](#).
- New command for setting headers: `\headings`; see [section 3](#).
- Updated theorem-like and proof environments using `ntheorem` package; see [section 6](#).
- Fixed handling of appendices to adhere to SIAM style guidelines; see [section 10](#).
- Added `review` class option; see [section 2](#).
- Includes the following packages by default: `algorithm`, `amsmath`, `breakurl`, `cleveref`, `hypcap`, `hyperref`, `ifpdf`, `ntheorem`, `xcolor`, and `xr-hyperref`. Adds the `lineno` package if the `review` class option is enabled.

**Appendix B. Special modifications.**

**B.1. Special macros.** In past versions, the SIAM standard L<sup>A</sup>T<sub>E</sub>X class defined the following macros: `\const`, `\diag`, `\grad`, `\Range`, `\rank`, and `\supp`. These are no longer declared; however, they can be redefined using the `\DeclareMathOperator` command from the `amsopn` package as demonstrated in [Example 10](#).

**B.2. Labeling objects sequentially.** SIAM recommends numbering objects by section number. However, if you prefer to number objects sequentially (e.g., Figure 5 would indicate the fifth figure appearing in the paper regardless of which section it is in), add the code from [Example 33](#) as documentclass options (e.g., `\documentclass[review,onefignum,onetabnum]{siamart251216}`).

#### Example 33: Labeling objects sequentially

```
oneeqnum % equations
onetabnum % tables
onefignum % figures
onealgnum % algorithms
onethmnum % theorem environments
```

**B.3. Appendices with no title.** The SIAM style manual [14] allows for an appendix that is numbered (by a letter) but has no title. We have a special command to create such an appendix: `\appendixnotitle`. This is equivalent to a `\section` command in the appendix except that it has no arguments.

**B.4. Changing the font of algorithm titles.** Currently, the algorithm title font matches the figure and table title fonts. To make the algorithm title font appear as small caps, insert the code in [Example 34](#) into the preamble.

#### Example 34: Changing algorithm title font

```
\makeatletter
\renewcommand{\ALG@name}{\sc Algorithm}
\makeatother
```

**B.5. Changing title of proof.** To change the title of a proof to, say, “Proof of main theorem,” simply use an optional argument as shown in [Example 35](#).

#### Example 35: Altered proof environment

```
\begin{proof}[Proof of main theorem]
We now show ...
\end{proof}
```

**B.6. Special instructions for PDF bookmarks.** Section titles are automatically inserted into the table of contents and therefore used as PDF bookmarks. Some special symbols may not correctly render. In that case, you can define alternate text as shown in [Example 36](#), using the `\texorpdfstring` command provided by the `hypertex` package. In this example, the default bookmark would be “Discussion of  $Z = X \cup Y$ ”; instead, it is replaced by “Discussion of  $Z = X \cup Y$ .”

#### Example 36: PDF bookmarks for symbols in section titles

```
\section{Discussion of \texorpdfstring{{\bfmath$Z=X \cup Y$}}{Z = X
union Y}}
```

## REFERENCES

- [1] AMERICAN MATHEMATICAL SOCIETY, *User’s guide for the `amsmath` package (version 2.0)*, 2002, <http://ftp.ams.org/pub/tex/doc/amsmath/amslatex.pdf> (accessed 2015-07-30).

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- [4] P. DAWKINS, *Paul's online math notes: Calculus I — notes*, <http://tutorial.math.lamar.edu/Classes/CalcI/MeanValueTheorem.aspx> (accessed 2015-07-08).
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- [13] D. E. WOESSNER, S. ZHANG, M. E. MERRITT, AND A. D. SHERRY, *Numerical solution of the Bloch equations provides insights into the optimum design of PARACEST agents for MRI*, Magnetic Resonance in Medicine, 53 (2005), pp. 790–799, <https://doi.org/10.1002/mrm.20408>, <https://www.ncbi.nlm.nih.gov/pubmed/15799055>.
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